NO: 2
2-2026

DETERMINATION OF CURRENT IN CIRCUIT USING MESH ANALYSIS

AIM:

To determine the current in circuit using mesh analysis both theortically and pratically for the given DC Circuit.

APPARATUS REQUIRED:

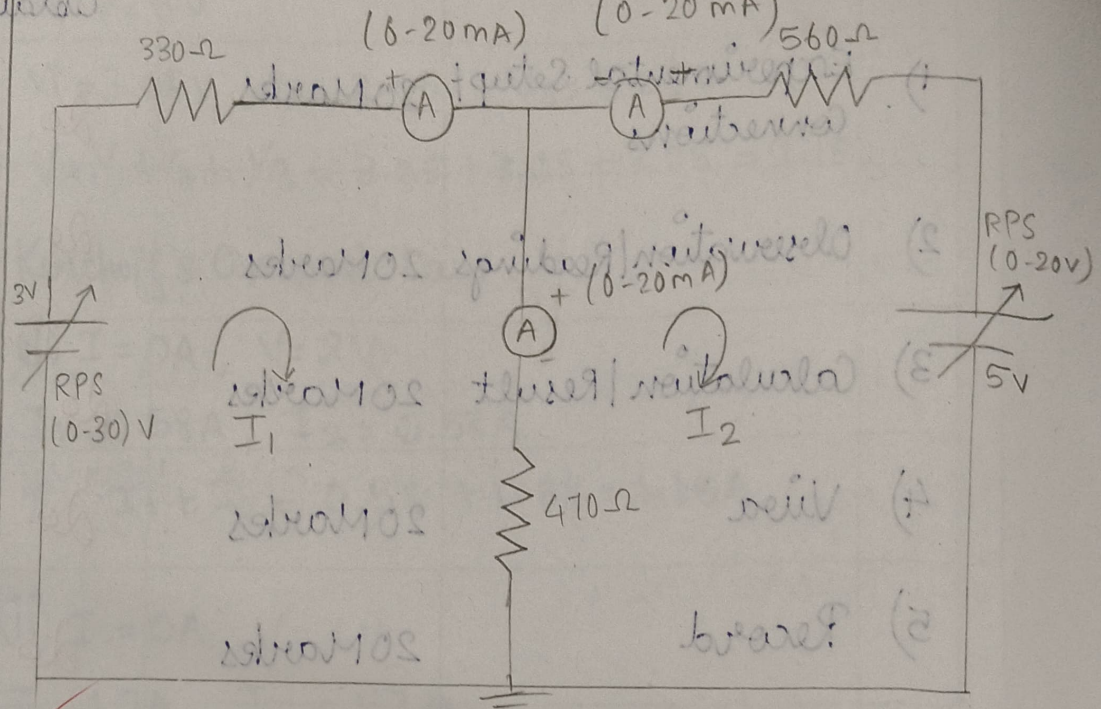
S.NO	APPARATUS	SPECIFICATION	QUANTITY
1)	Regulated Power Supply (RPS)	(0-30 V)	1
2)	Ammeter	(0-10mA) MC	3
3)	Resistors	330Ω, 470Ω, 560Ω	Each 1
4)	Bread Board	-	1

PROCEDURE:

1) Give connections as per the circuit diagram

2) Switch ON the Supply, Vary the RPS (Regulated Power Supply) and Set a particular input voltage.

CIRCUIT DIAGRAM:



Mesh Circuit

3) Note down the readings of ammeters and voltmeters and tabulate them.

4) Vary the RPS for different input voltages and note down the readings of all the meters.

5) Reduce the RPS to its minimum value and switch ~~of~~ OFF the supply.

6) Using the tabulated values, verify Kirchoff's laws pratically and verify it theoretically.

TABULAR COLUMN:

Parameters	Theoretical	Pratical
I_1	1.2	1.2
I_2	4.2	4.23
$I_1 - I_2$	5.4	5.4

O/P verified

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Theoretical Calculation

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$R_{11} = 330\Omega + 470\Omega = 800\Omega$

$R_{12} = 470\Omega$, $R_{21} = 470\Omega$

$R_{22} = 560 + 470 = 1030\Omega$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$\Delta = 800 \times 1030 - (470)^2$

$= 603100.$

$$\Delta_1 = \begin{vmatrix} 3 & -470 \\ -5 & 1030 \end{vmatrix}$$

$= 3090 - 2350 = 740$

$$\Delta_2 = \begin{vmatrix} 800 & 3 \\ -470 & -5 \end{vmatrix} = -4000 + 1410 = 2590$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{603100} = 1.2\,mA$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-2590}{603100} = -4.2\,mA$$

$I_1 - I_2 = 1.2 - (-4.2) = 5.4\,mA$

MARKS SPILT UP:

S.No	Description	Marks Alloted	Marks Obtained
1)	Experiment / Setup / Connections	20 Marks	20
2)	Observation / Readings	20 Marks	20
3)	Calculation / Result	20 Marks	20
4)	Viva	20 Marks	20
5)	Record	20 Marks	20
		Total	100

RESULT:

10

Thus, The Mesh analysis is verified pratically and theoretically. The resultant currents for 3/5 V Supply are:

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a) The Current across 330 Ω resistor is 1.2 mA.

b) The Current across 560 Ω resistor is 4.23 mA.

c) The Current across 470 Ω resistor is 5.4 mA.